

Measuring Financial Subsidies to Chinese State-Owned Enterprises:

Machine-Learning Estimation

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Abstract

We estimate financial subsidies to Chinese A-share listed state-owned enterprises (SOEs) using the asset-return framework of Lucas, Garcia, and Solberg (2026). A strictly out-of-time machine-learning model predicts private-sector EBI/A, while a 5-nearest-neighbor diagnostic measures common support. From 2019–2025, the average subsidy rate is 1.45 percentage points, with annual estimates from 0.74 to 1.88 pp and 1.42 pp within common support. These magnitudes are broadly consistent with prior Chinese debt-side evidence. The 2025 subsidy amount is CNY 682.6 billion. The measure is an asset-return opportunity-cost estimate, not a budget cash transfer or a causal privatization effect.

Keywords: state-owned enterprises; financial subsidies; financial subsidy rate; EBI/A; machine learning; private-sector benchmark

1 Introduction

Government support to state-owned enterprises matters for resource allocation, public-sector opportunity costs, competition, and the interpretation of SOE profitability. The scale of Chinese state support is also a prominent policy question in sectors such as electric vehicles and solar panels, where subsidy concerns have shaped recent trade and competition debates (European Commission, 2024; OECD, 2026). Visible grants and tax preferences capture only part of that support because governments can also accept below-market returns on loans, guarantees, or equity. A measure that misses those financing concessions can understate the economic value of support; a measure that adds overlapping channels can overstate it. The central empirical question is therefore: how large are the financial subsidies received by Chinese listed SOEs? We answer this question by measuring the gap between each SOE’s realized asset return and the return that a comparable private firm would be expected to earn, and by translating that return gap into an annual subsidy amount.

Lucas, Garcia, and Solberg (2026) provide the economic starting point. They define a financial subsidy as the return concession the government accepts when financing an SOE and move the

measurement problem from individual debt and equity instruments to the return generated by the firm’s assets. Their accounting implementation uses earnings before interest over assets (EBI/A), which adds interest expense back to net income and is therefore less mechanically sensitive to capital structure than ROA or ROE. The subsidy rate equals the private-sector asset-return benchmark minus the SOE’s realized asset return; multiplying that rate by book assets gives the annual subsidy amount.

The arithmetic is easy; the private-sector benchmark is not. Average SOE–private return gaps mix differences in size, leverage, asset structure, liquidity, investment, market position, and industry composition. Constructing comparable private-sector benchmarks consistently for more than a thousand heterogeneous listed SOEs is difficult if each firm must be matched case by case. A benchmark for a given forecast year must also exclude outcomes realized in that same year, or future information leaks into the comparison. Some large or unusual SOEs may have no close private counterpart, so a model can produce a prediction even when the underlying comparison rests on extrapolation.

We implement that framework for the Chinese A-share market in three ways. First, we construct a firm-specific conditional private-sector benchmark from the historical performance of the eligible private-firm sample, using a gradient boosting model (GBM) as the main specification and Ridge regression and random forests as alternative functional forms. Second, every forecast is strictly out of time: the model, tuning inputs, and preprocessing for year τ use only private-firm outcomes realized before τ . Third, a firm-level 5-nearest-neighbor (5-NN) diagnostic classifies each SOE observation as Good support, Boundary support, or Extrapolation, making the strength of the private-sector comparison explicit rather than implicit.

The estimates are economically large and stable for the full market. From 2019–2025, the GBM financial subsidy rate averages 1.45 percentage points (pp), with a cluster-bootstrap 95% confidence interval of [1.25, 1.64]. The rate is 1.42 pp within common support and 1.41 pp among persistent-good firms, defined as firms classified as Good support in at least 80% of their observed years. The CSI indices provide a useful size-based heterogeneity comparison: the CSI 300 contains the largest and most liquid A-share firms, the CSI 500 covers the next tier of smaller firms outside the CSI 300, and the CSI 1000 covers a still smaller segment outside the CSI 800. CSI 500 and CSI 1000 SOEs average 1.34 pp and 1.39 pp, while the CSI 300 estimate is -0.07 pp and statistically indistinguishable from zero. This comparison is informative because the largest SOEs have substantially weaker overlap with private-firm benchmarks. In 2025, firm-level rates multiplied by book assets sum to CNY 682.6 billion; Good and Boundary firms account for 83.6% of that amount.

The magnitude is economically meaningful and broadly consistent with prior Chinese financing-advantage estimates. Our annual asset-return gaps range from 0.74 to 1.88 pp, with a 1.45 pp average. Debt-side studies report a 0.43 pp effective-interest-rate advantage for listed SOEs after controls (Jurzyk and Ruane, 2021), average bond-spread advantages of 0.81 pp for local SOEs and 1.75 pp for central SOEs (Zhang and Wang, 2020), and implicit-guarantee premia of 0.72–1.66 pp across SOE bond ratings (Zhang, Li, and Tian, 2022). The estimands differ, but they place our

estimate in the same broad order of magnitude.

The empirical contribution is therefore a scalable implementation of an existing economic definition. The asset-return framework supplies the measurement object; the strictly out-of-time conditional benchmark supplies a firm-specific private-sector counterfactual; and the common-support diagnostic shows which estimates are anchored in observed private-firm comparables. Together, they scale the asset-return comparison to a large and heterogeneous panel without changing the underlying “return difference $\times$ assets” logic.

2 Background and Related Literature

2.1 Traditional Approaches to Measuring SOE Support

Visible support—grants, tax preferences, and related fiscal measures—can often be identified in government or corporate accounts (Wu, 2025; Huang, 2023; Bradshaw et al., 2019). Their visibility does not remove measurement problems: subsidy applications can contain information frictions, and the relation between SOE profit remittances and government subsidies can complicate a simple net-transfer interpretation (Jin, 2018; Wu, 2025).

Financing support is harder to price because it often works through implicit guarantees, below-market loans, or preferential access to credit rather than an explicit budget line. Expected government support can improve an SOE’s perceived creditworthiness and lower financing costs, the mechanism emphasized by the soft-budget-constraint literature (Dong, 2003). Evidence for Chinese SOEs is consistent with financing advantages linked to state ownership and state backing (Ge et al., 2020; Maurel et al., 2020; Jin et al., 2022). Broader approaches increasingly consider subsidies, equity injections, and government or state-linked lending together rather than treating each instrument in isolation (Herrera Dappe et al., 2024).

Following Lucas et al. (2026), we use the term *financial subsidy* to describe return concessions associated with debt and equity financing, including below-market government loans, underpriced debt guarantees, and government equity that earns less than a comparable private investor would require. Grants, tax preferences, price support, and underpriced land are non-financial subsidies in that framework. Both can transfer value to an SOE, but the empirical object below is the financial subsidy measured from asset returns.

2.2 Why Channel-by-Channel Measurement Can Go Wrong

Channel-by-channel measurement is intuitive because each item can be named and valued separately. The total can still be wrong. A fiscal-account measure can miss implicit guarantees, low-cost financing, and preferential credit when those forms of support have no observable market price (Maurel, 2020). Adding separate valuations can also double count benefits when grants, tax preferences, loans, equity injections, and guarantees overlap or substitute for one another (Huang, 2023; Li, 2022; Herrera Dappe et al., 2024).

Debt and equity support also interact. A below-market government loan reduces interest expense, which can raise the residual return to equity; an equity concession can alter the value of debt support in the opposite direction. Measuring the debt concession and the equity concession independently can therefore miss offsets within the capital structure (Lucas et al., 2026).

ROA and ROE add another mechanical problem because net income is measured after interest expense. Two firms with similar underlying asset performance can report different ROA or ROE solely because they use different leverage. ROE also does not assign an explicit opportunity cost to government equity. An asset-side return measure reduces those capital-structure distortions and provides one scale for debt and equity support.

2.3 From Asset Returns to Financial Subsidies

The Lucas et al. (2026) approach asks how much lower the return on an SOE’s assets is than the return on comparable private-sector assets. Assets are financed by debt and equity, and the cash flows generated by those assets are shared across both claims. Moving the comparison to the asset side therefore reduces omissions and double counting that can arise when financing instruments are priced one by one.

With accounting data, Lucas et al. recommend EBI/A as a practical proxy for the return on assets. In our data,

$$EBI = NI + Interest, \quad (1)$$

and

$$\frac{EBI}{A} = \frac{NI + Interest}{Assets}. \quad (2)$$

EBI measures income generated before interest is paid to lenders, and EBI/A scales that income by the asset base. Adding interest expense back to net income makes EBI/A less mechanically sensitive to debt use than a net-income-based return measure.

Lucas et al. write the annual financial subsidy as

$$Financial\ Subsidy = [E(r_{A,Private}) - E(r_{A,SOE})] \times A_{SOE}. \quad (3)$$

Here, $E(r_{A,Private})$ is the expected asset return of a comparable private-sector firm—the private-sector benchmark— $E(r_{A,SOE})$ is the SOE’s asset return, and A_{SOE} is the SOE’s book total assets. The bracketed difference is the financial subsidy rate. If the private-sector benchmark is 5% and the SOE earns 3.5%, the rate is 1.5 pp; CNY 10 billion of book assets then implies an annual financial subsidy of CNY 150 million.

We retain Lucas et al.’s terminology and rate-times-assets conversion. Their framework supplies the economic measurement object; the comparisons below use Chinese evidence to benchmark the magnitude of our estimates.

2.4 Constructing a Scalable Conditional Private-Sector Benchmark

The main empirical choice is the private-sector benchmark, $E(r_{A,Private})$. Lucas et al. (2026) discuss peer-based benchmarks and asset-pricing benchmarks such as the CAPM. Peer comparisons stay close to realized operating data but depend on finding sufficiently similar firms. Asset-pricing benchmarks avoid hand-picking peers but depend on assumptions about asset beta, the risk-free rate, and the market risk premium.

For the A-share cross-section, we operationalize the private-sector comparison as a conditional prediction problem. For forecast year τ , historical private-firm data estimate how next-period EBI/A varies with observable financial characteristics and industry. Applying that relationship to an SOE’s lagged characteristics yields its predicted private-sector EBI/A. The construction preserves the Lucas et al. measurement object while providing one common, scalable rule for constructing firm-specific benchmarks across SOEs and years.

Two safeguards address problems that a flexible prediction model cannot solve by itself. The expanding training window excludes all outcomes from the forecast year, eliminating direct time leakage. The 5-NN support diagnostic separately asks whether the predicted benchmark lies near observed private firms in covariate space. A prediction can be numerically precise yet economically weak if it relies on extrapolation; support status makes that distinction visible.

3 Data

3.1 Data Sources

The data come from the Wind Financial Terminal and were extracted in August 2026. They cover annual financial statements for all Chinese A-share listed firms from 2014–2024 and support forecasts for 2019–2025.

Data	Source	Coverage
Financial statements for all A-share listed firms	Wind Financial Terminal	11-year panel, 5,535 firms
Index constituent entry/exit records	Wind Financial Terminal	CSI 300 / CSI 500 / CSI 1000, 2019–2025

Financial characteristics in year t predict EBI/A in year $t + 1$; for example, 2024 characteristics predict 2025 returns. The feature period is 2014–2024 and the forecast period is 2019–2025. We begin forecasting in 2019 so that the initial model is estimated on four prior outcome years, 2015–2018, generated from predictor data in 2014–2017. Each subsequent forecast adds another year of realized private-firm outcomes to the expanding training window, while retaining the requirement that all training outcomes precede the forecast year.

3.2 Sample Construction and Screening

Table 1 reports the sample-construction process. The raw data contain 5,535 firms and 60,896 firm-year observations. After excluding financial firms, observations with missing returns, extreme returns, and unclassified ownership, the private-firm training pool contains 34,462 firm-year observations from 3,471 firms. Requiring complete predictor data leaves 27,722 firm-years from 3,471 private firms before the final screens.

We apply three additional screens to remove unusually distressed or extreme firms:

1. Special Treatment (ST) stocks are excluded (firm names containing “ST”; 127 private firms and 58 SOEs). ST status is an exchange designation for listed firms facing abnormal financial or operating conditions. We exclude these firms because severe distress can generate unusually extreme accounting returns and make them poor benchmarks for ordinary operating firms.
2. Firms with negative book equity are excluded ($\text{FinancialLeverage} > 1$).
3. Extreme profit/loss observations are excluded ($|EBI/A| > 0.5$).

The final private-firm training sample contains 26,464 firm-years from 3,344 firms. The 2025 forecast sample contains 1,336 A-share SOEs, including 436 central SOEs and 900 local SOEs.

Table 1: Sample Construction

Step	Operation	Firms	Firm-years	Firms removed	Firm-years removed
0	Parse 11 annual Wind files	5,535	60,896	—	—
1	Remove observations without $t + 1$ total assets	5,535	57,701	0	3,195
2	Backfill missing 2014 ownership (100% recovered)	5,535	57,701	0	0
3	Exclude financial industries	5,416	56,392	119	1,309
4	Remove missing EBI/A	5,416	54,741	0	1,651
5	Initial broad screen: EBI/A outside $(-1, +1)$	5,416	54,671	0	70
6	Remove unclassified ownership	5,416	54,671	0	0
7	Keep private firms only (training pool)	3,471	34,462	1,945	20,209
8	Complete cases for predictors	3,471	27,722	0	6,740

Step 5 is a broad validity screen in the raw-data cleaning stage. After Step 8, the final screens remove ST firms, observations with $\text{FinancialLeverage} > 1$, and observations with $|EBI/A| > 0.5$. The last rule removes an additional 129 firm-years. The final training sample remains 26,464 firm-years from 3,344 private firms.

The available data do not permit reliable additional screens based on listing age or very low share prices. Section 4.5 discusses the resulting measurement limitations.

3.3 Variable Definitions

The target variable is next-year EBI/A:

$$\frac{EBI}{A}_{t+1} = \frac{\text{Net income attributable to parent}_{t+1} + \text{Interest expense}_{t+1}}{\text{Total assets}_{t+1}}. \quad (4)$$

The numerator is net income attributable to parent shareholders plus interest expense, a post-tax, pre-interest proxy for the return on assets. Adding back interest expense reduces the mechanical effect of capital structure and debt-financing costs. Because taxes remain in the numerator, the measure is not EBIT.

Two accounting points matter. First, the numerator is attributable to parent shareholders while total assets are consolidated, so the accounting scopes do not perfectly match. Second, tax differences can move the measured subsidy rate mechanically. If SOEs face lower effective tax rates than comparable private firms, post-tax SOE EBI/A rises relative to the private benchmark and the measured shortfall shrinks; all else equal, that difference biases the headline 1.45 pp subsidy rate downward, toward zero, relative to a pre-tax comparison.

The predictors are measured in year t and include nine financial ratios plus industry fixed effects. MarketShare and HHI use total A-share listed-firm revenue within each industry-year as the denominator rather than revenue within an index subsample.

Table 2: Predictor Definitions

Feature	Definition	Economic meaning
FirmSize	$\ln(\text{total assets})$	Firm size
AssetTurnover	Revenue / total assets	Asset-use efficiency
FinancialLeverage	Total liabilities / total assets	Leverage
FixedAssetsRatio	Net fixed assets / total assets	Asset intensity
CurrentRatio	Current assets / current liabilities	Short-term liquidity
CapexRatio	Capital expenditure / total assets	Investment intensity
WorkingCapitalRatio	(Current assets - current liabilities) / total assets	Working-capital position
MarketShare	Revenue / listed-industry revenue	Listed-market industry share
HHI	$\sum \text{MarketShare}^2$ within industry-year	Listed-market industry concentration

3.4 Ownership Classification

SOEs are central SOEs plus local SOEs; private firms are firms classified as private enterprises. Public enterprises, foreign firms, collective firms, and other ownership categories are excluded from the benchmark sample.

The Wind ownership field is a static latest-date snapshot rather than historical ownership status measured in year t . Ownership changes can therefore classify earlier observations with hindsight. The data audit identifies about 95 firm-year observations associated with ownership switchers, limiting the likely scope of the problem but not eliminating it.

4 Method

4.1 Private-Sector Conditional Prediction Framework

For forecast year τ , the model uses only private-firm observations whose outcome years are strictly earlier than τ . Define the historical private-firm information set as

$$\mathcal{D}_{<\tau}^P = \{(X_{j,s-1}, Y_{j,s}) : Private_j = 1, s < \tau\}, \quad (5)$$

where Y is raw, non-winsorized EBI/A. We estimate $\hat{m}_\tau^P(x)$ on that information set and apply it to SOE i 's characteristics in $\tau - 1$:

$$\hat{Y}_{i,\tau}^P = \hat{m}_\tau^P(X_{i,\tau-1}). \quad (6)$$

Following Lucas et al. (2026), the financial subsidy rate is

$$\widehat{FSR}_{i,\tau} = \hat{Y}_{i,\tau}^P - Y_{i,\tau}^{SOE}. \quad (7)$$

Both terms are raw EBI/A. A positive $\widehat{FSR}$ means that realized SOE EBI/A falls below the out-of-time private-sector benchmark; a negative rate means that realized EBI/A exceeds it. The annual financial subsidy is

$$\widehat{FS}_{i,\tau} = \widehat{FSR}_{i,\tau} \times A_{i,\tau}^{SOE}, \quad (8)$$

where $A_{i,\tau}^{SOE}$ is book total assets.

4.2 Model Specifications

Private-firm observations estimate EBI/A_{t+1} as a function of the nine financial predictors and industry fixed effects. We use three functional forms:

Model	Hyperparameters	Role
Gradient Boosting (GBM)	$n = 200$, depth= 4, lr= 0.05, min_samples_leaf= 10	Main model
Ridge regression	$\alpha = 10$	Linear robustness benchmark
Random forest (RF)	$n = 300$, depth= 8, min_samples_leaf= 10	Nonlinear robustness benchmark

Model choice follows an ex ante rule based on out-of-time fit and calibration in private firms, not on SOE results. With the raw target, pooled out-of-fold (OOF) R^2 is 0.239 for GBM, 0.172 for Ridge, and 0.206 for RF. Calibration slopes are 0.98 for GBM, 0.97 for Ridge, and 1.39 for RF. GBM offers the best overall balance of explanatory power, RMSE, and calibration and serves as the main model.

The nine numeric predictors are winsorized at P5–P95, median-imputed, standardized with `StandardScaler`, and combined with one-hot indicators for second-level industries. The main

specification does not winsorize the outcome. Training labels and realized SOE EBI/A both use the raw outcome after sample screening. Each forecast year’s preprocessing parameters come only from the historical private-firm training sample available at that date and are then applied to SOEs.

4.3 Cross-Validation and Statistical Inference

Five-fold `GroupKFold` cross-validation by `firm_id` evaluates private-firm predictive performance. All years for the same firm remain in one fold, preventing the same firm from appearing in both training and validation samples.

The counterfactual benchmark uses an expanding window subject to

$$\max(\text{OutcomeYear}_{\text{train}}) < \text{ForecastYear}. \quad (9)$$

The 2025 benchmark, for example, uses private-firm outcomes realized no later than 2024; private-firm outcomes from 2025 do not enter training.

Because observations for the same firm can be correlated across years, annual standard errors are clustered at the firm level. For the annual mean financial subsidy rate,

$$SE = \frac{sd(\text{firm-level means})}{\sqrt{n_{\text{firms}}}}. \quad (10)$$

Multiyear, common-support, and persistent-good results use 2,000 firm-level cluster-bootstrap draws (seed=42) applied to fitted firm-year subsidy rates. The bootstrap preserves within-firm serial dependence. It measures sampling uncertainty conditional on the fitted prediction function and does not include uncertainty from retraining the first-stage model.

4.4 Common-Support Diagnostics

A credible conditional benchmark also requires overlap between SOEs and the private-firm covariate distribution. We use four diagnostics:

1. **Standardized mean differences (SMDs):** $(\bar{X}_{SOE} - \bar{X}_{\text{private}})/\sigma_{\text{pooled}}$. We treat $|SMD| > 0.25$ as economically nontrivial.
2. **Propensity scores:** five-fold OOF probabilities from a logistic model that distinguishes SOEs from private firms summarize covariate separability and overlap.
3. **Multivariate 5-NN distance:** for forecast year T , preprocessing uses only historical private-firm observations with outcome years below T . To prevent repeated observations from one private firm from acting as multiple neighbors, define

$$d_j(x) = \min_{s \in j} \|z(x) - z_{j,s}\|_2. \quad (11)$$

Each private firm contributes only its minimum historical distance. $d^{(5)}(x)$ is the fifth-smallest

$d_j(x)$ across distinct private firms. For the private-firm reference distribution, all observations of the query firm are removed before calculating the same statistic. Forecast-year-specific $P90_T$ and $P95_T$ thresholds define Good support ($d^{(5)} \leq P90_T$), Boundary support ($P90_T < d^{(5)} \leq P95_T$), and Extrapolation ($d^{(5)} > P95_T$). From 2019 to 2025, $P90_T$ declines from 1.532 to 1.244 and $P95_T$ from 1.695 to 1.404 as the historical private-firm pool expands.

4. **Mahalanobis distance:** distance on the first five principal components provides a cross-check.

Appendix Figure A1 shows the largest visible one-dimensional mismatch: firm size. The private-firm distribution is centered near $\ln(\text{assets}) \approx 21.7$, the full SOE distribution near 23.0, and the CSI 300 SOE distribution near 25.7. About 93% of CSI 300 SOEs exceed the private-firm P95 in firm size; the shares are about 63% for CSI 500 and 25–28% for CSI 1000 and the full SOE sample. The CSI 300 size SMD is +3.19. Because a one-variable P95 exceedance does not imply multivariate support failure, the final interpolation/extrapolation classification uses the 5-NN measure.

4.5 Interpretation and Measurement Constraints

The design has four limits that shape interpretation. First, $\widehat{FSR}$ is a benchmark return difference, not a causal treatment effect of changing ownership. Multiplying the rate by book assets produces an annual asset-return opportunity-cost measure; it does not decompose the estimate into cheap credit, guarantees, equity-return concessions, grants, or tax preferences. Mechanism and incidence require additional identification.

Second, common support is limited by the observed information set. The 5-NN diagnostic reveals where SOEs resemble observed private firms, but comparability can use only the financial and industry variables in the data. Large CSI 300 SOEs overlap less with private firms and their estimates are more sensitive to model choice. Product-market, geographic, governance, and policy-mandate variables could improve benchmarks for those firms.

Third, the data and accounting definitions impose measurement constraints. The ownership field is a static latest-date snapshot; listing-date, listing-status, and stock-price fields cannot support additional screens; and EBI uses parent-attributable net income with consolidated total assets. Book returns may also reflect asset revaluations, impairments, and industry-specific accounting practices. The post-tax EBI definition can bias the measured return gap toward zero when SOEs have lower effective tax rates than comparable private firms.

Fourth, the bootstrap conditions on the fitted prediction function and therefore omits first-stage retraining uncertainty. External validity is also limited to A-share listed firms. Listed SOEs differ from unlisted SOEs in size, governance, financing access, and disclosure, so the estimates should not be mechanically generalized to the entire Chinese state-owned sector.

5 Results

5.1 Model Fit and Calibration

Table 3 reports pooled out-of-time predictive performance for private firms under the raw-outcome specification. GBM has the highest out-of-fold (OOF) R^2 and the lowest RMSE, and its calibration slope of 0.98 is close to one. Ridge is also well calibrated but explains less variation. RF has slightly lower MAE than GBM but a calibration slope of 1.39, indicating stronger shrinkage toward the mean.

Table 3: Out-of-Time Prediction Performance for Private-Firm Raw EBI/A

Model	OOF R^2	OOF RMSE	OOF MAE	Calibration β
Ridge regression	+0.172	0.0671	0.0432	0.97
Random forest	+0.206	0.0657	0.0406	1.39
Gradient boosting	+0.239	0.0644	0.0410	0.98

Note: Both training targets and validation outcomes are raw EBI/A. The nine predictors are winsorized at training-sample P5–P95. The calibration regression is $\text{Actual} = \alpha + \beta \times \text{Predicted}$.

5.2 All A-Share SOEs: An Average Financial Subsidy Rate of About 1.5 pp

Realized EBI/A for all A-share SOEs remains below the private-sector conditional benchmark throughout 2019–2025. Table 4 reports annual estimates, and Figure 1 plots the GBM estimates and confidence intervals.

Table 4: Annual Financial Subsidy Rates for All A-Share SOEs

Year	Firms	GBM subsidy rate	95% CI	Ridge rate	RF rate
2019	1,188	+1.32 pp*	[+0.99, +1.64]	+0.95 pp	+1.65 pp
2020	1,241	+1.82 pp*	[+1.47, +2.16]	+1.32 pp	+1.95 pp
2021	1,301	+0.74 pp*	[+0.44, +1.05]	+0.39 pp	+1.11 pp
2022	1,331	+1.40 pp*	[+1.11, +1.70]	+0.90 pp	+1.57 pp
2023	1,342	+1.46 pp*	[+1.20, +1.73]	+0.87 pp	+1.64 pp
2024	1,339	+1.88 pp*	[+1.60, +2.16]	+1.28 pp	+2.14 pp
2025	1,336	+1.53 pp*	[+1.27, +1.78]	+1.04 pp	+1.85 pp

Note: * indicates that the firm-clustered 95% confidence interval excludes zero.

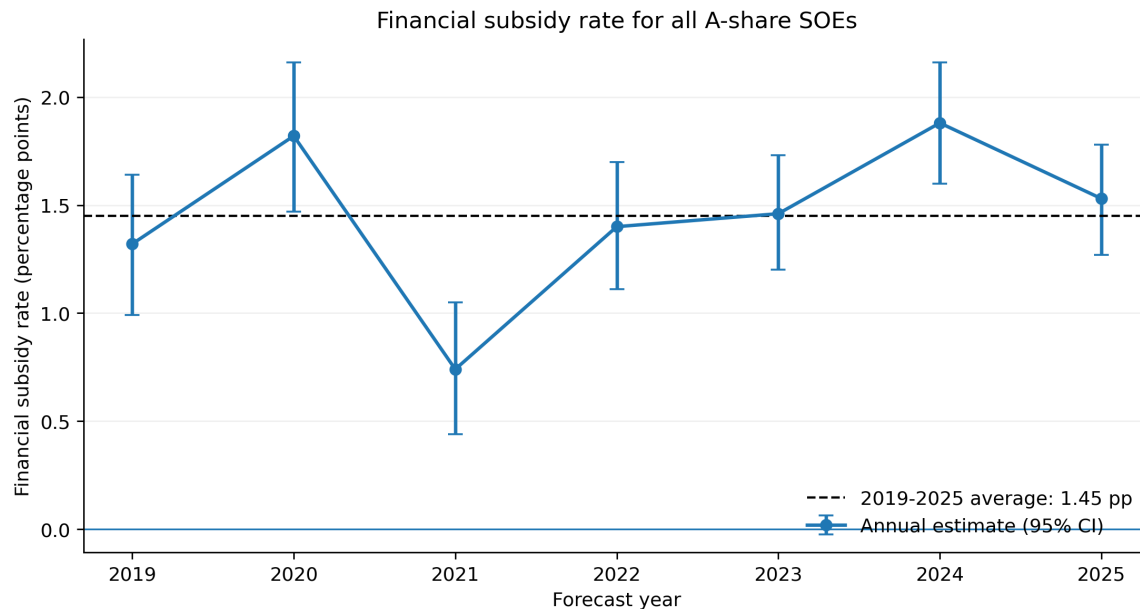


Figure 1: Annual financial subsidy rates for all A-share SOEs. Points are GBM estimates; vertical bars are firm-clustered 95% confidence intervals. The dashed line marks the 2019–2025 average of 1.45 pp.

The GBM rate is positive in every year, ranging from +0.74 pp to +1.88 pp, and every annual 95% confidence interval excludes zero. Pooling all years gives a firm-year average of +1.45 pp (cluster-bootstrap 95% CI: [+1.25, +1.64]). Seven consecutive positive estimates make a single macroeconomic episode an unlikely explanation for the full-market result.

Observable financial characteristics and industry composition do not eliminate the return gap. Ridge and RF produce multiyear full-sample rates of +0.97 pp and +1.70 pp, both significantly positive. Nonlinearities, interactions, and model class change the size of the estimate but not its full-market sign.

5.3 Magnitude Relative to Earlier Chinese Estimates

Same-year comparisons provide a second scale benchmark. In 2019, our subsidy rate is 1.32 pp and the corresponding amount is CNY 555.7 billion, about 0.56% of final GDP reported by the National Bureau of Statistics of China (2020). DiPippo, Mazzocco, and Kennedy (2022) estimate average SOE borrowing advantages of 1.4 pp on corporate bonds and 0.5 pp on bank loans; the same report estimates China’s below-market credit to be at least CNY 509 billion, or 0.52% of GDP, in 2019. In 2023, our CNY 637.5 billion estimate is about 0.49% of revised GDP (National Bureau of Statistics of China, 2024), while Garcia-Macia, Kothari, and Tao (2025) estimate subsidized credit for favored sectors at 0.4% of GDP.

The comparisons establish order of magnitude rather than equality of estimands. Earlier estimates price liability-side borrowing advantages or industrial-policy credit, whereas our measure is an asset-side opportunity-cost gap for listed SOEs.

5.4 Index Groups: Positive Subsidy Rates for CSI 500/1000, Near Zero for CSI 300

Positive subsidy rates are concentrated in CSI 500 and CSI 1000 SOEs, while the multiyear CSI 300 rate is close to zero. Table 5 reports annual estimates, and Figure 2 compares the three index paths.

Table 5: Index-Group Heterogeneity: Annual GBM Financial Subsidy Rates (pp)

Sample	2019	2020	2021	2022	2023	2024	2025
CSI 300 SOEs	+0.43	+1.09*	−0.53	−0.83	−0.33	−0.24	−0.04
CSI 500 SOEs	+1.12*	+1.58*	+0.41	+1.25*	+1.94*	+1.88*	+1.22*
CSI 1000 SOEs	+1.96*	+1.67*	+0.45	+1.49*	+1.67*	+1.28*	+1.22*
Non-index SOEs	—	—	—	—	—	—	+2.03*

Note: * indicates that the 95% CI excludes zero. Non-index SOEs are reported only for 2025.

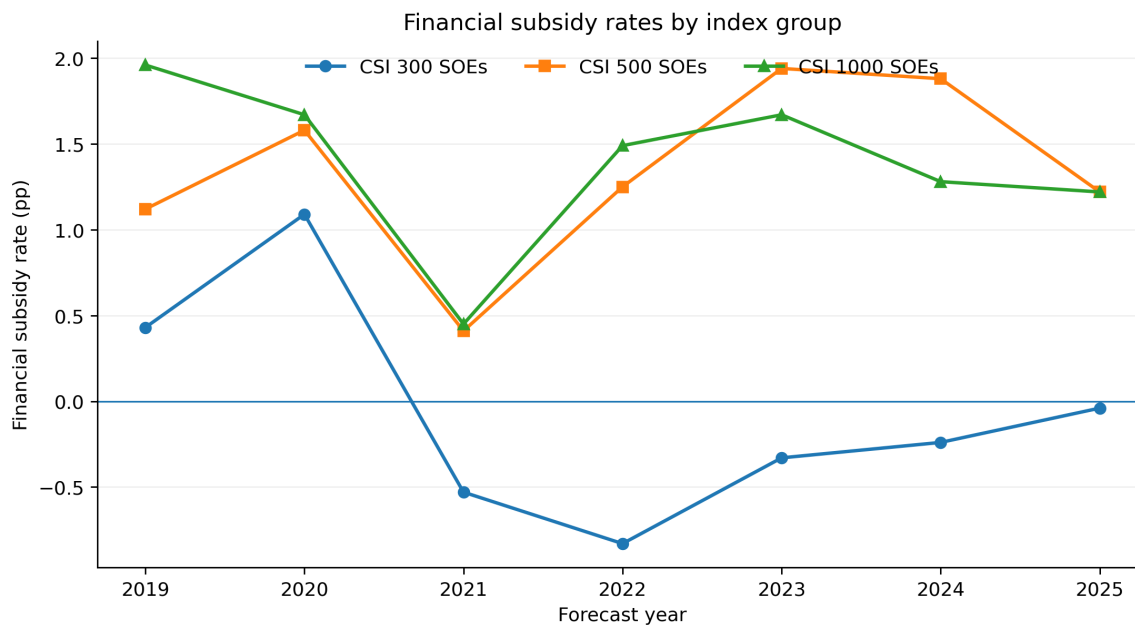


Figure 2: Annual financial subsidy rates by index group (GBM, raw outcome).

Under GBM, the multiyear rates are +1.34 pp for CSI 500 SOEs and +1.39 pp for CSI 1000 SOEs, with 95% confidence intervals excluding zero. The CSI 300 average is -0.07 pp and statistically indistinguishable from zero. CSI 500 and CSI 1000 remain positive in most years, with a common weakening in 2021; CSI 300 is near zero or negative from 2021 onward.

The full-market rate is therefore not evenly distributed across listed SOEs. CSI 300 firms have weaker private-sector overlap and greater sensitivity to model class, consistent with their scale, market position, industry composition, and policy roles. CSI 500 and CSI 1000 estimates are more stable across years and support restrictions.

5.5 Industry Heterogeneity: Large Differences and Shifting Rankings

Financial subsidy rates vary substantially across industries, and industry rankings change over time. The 2019–2025 averages place Steel, Textiles & Apparel, Health Care Equipment & Services, Media, and Home Appliances among the higher-rate industries. Transportation and Coal are near zero or slightly negative on a multiyear basis, yet Coal ranks among the high-rate industries in the 2025 cross-section. The 1.45 pp market average therefore combines persistent industry differences with substantial time variation.

Capital-intensive and cyclical industries with greater policy involvement, such as Steel, remain relatively high over the multiyear sample. Coal moves sharply over time, showing how temporary industry conditions can change annual rankings even when longer-run institutional differences persist.

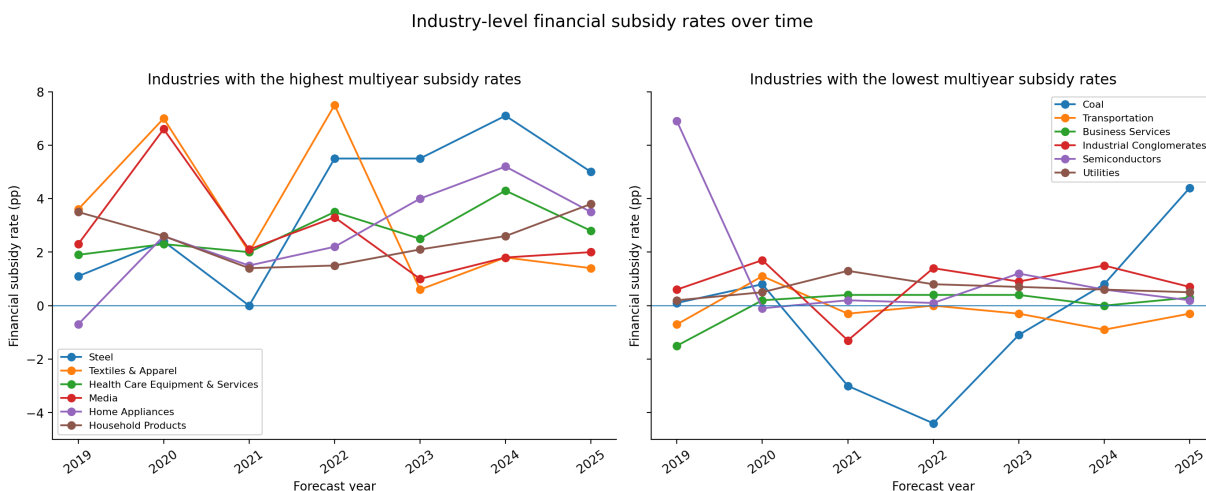


Figure 3: Industry-level financial subsidy rates over time. The left panel shows industries with the highest 2019–2025 average rates; the right panel shows the lowest.

Figure 4 compresses the same variation into industry-year cells. Positive and negative rates often persist for several years, but some industries move sharply across the sample period.

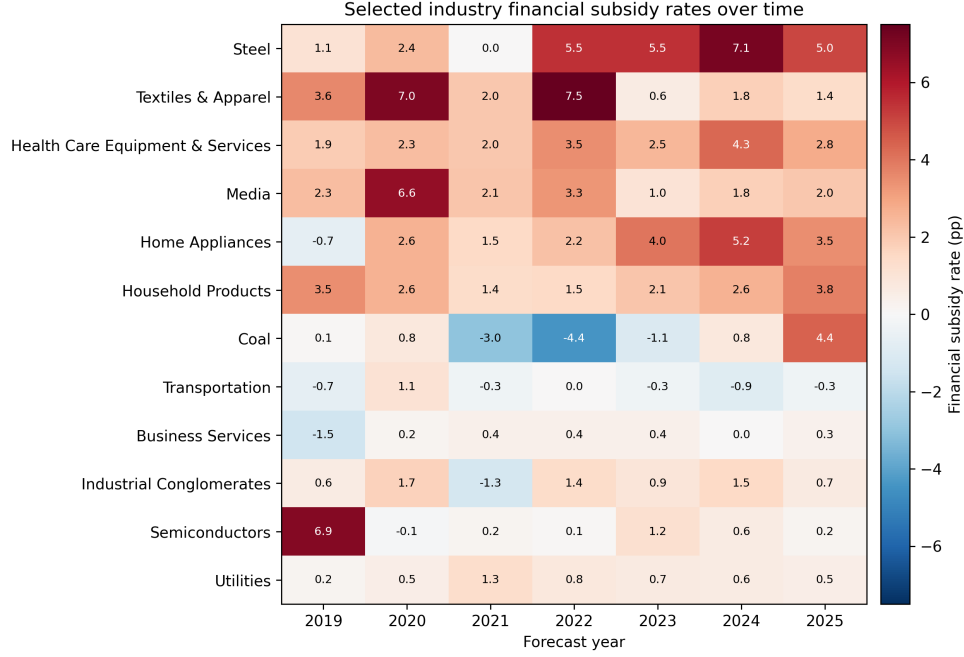


Figure 4: Selected industry financial subsidy rates by year. The industries correspond to the high- and low-subsidy groups highlighted in Figure 3. Cell values are percentage points.

Table 6: Industries with the Highest and Lowest Financial Subsidy Rates in 2025

Rank	Highest-rate industry	Mean rate	<i>n</i>	Lowest-rate industry	Mean rate	<i>n</i>
1	Steel	+5.01 pp	25	Telecommunication Services	-1.03 pp	4
2	Coal	+4.43 pp	29	Transportation	-0.27 pp	74
3	Real Estate	+4.42 pp	52	Industrial Conglomerates	-0.10 pp	22
4	Household Products	+3.84 pp	3	Software Services	+0.08 pp	47
5	Home Appliances	+3.54 pp	13	Semiconductors	+0.20 pp	26
6	Building Materials	+2.98 pp	20	Utilities	+0.56 pp	98

5.6 Central versus Local SOEs: No Significant Difference in the Average Subsidy Rate

Central and local SOEs have similar average subsidy rates, so administrative ownership does not explain the main full-sample heterogeneity. Table 7 reports the 2025 split.

Table 7: Central versus Local SOEs in 2025 (GBM)

Group	Firms	Actual EBI/A	Predicted EBI/A	Subsidy rate	SE	95% CI
Central SOEs	436	+2.22%	+3.79%	+1.58 pp	0.22 pp	[+1.15, +2.00]
Local SOEs	900	+1.68%	+3.19%	+1.50 pp	0.16 pp	[+1.19, +1.82]

The 2025 rates are +1.58 pp for central SOEs and +1.50 pp for local SOEs. The difference is about +0.07 pp (clustered test $p = 0.79$). Pooling 2019–2025 gives +1.40 pp for central SOEs

and +1.47 pp for local SOEs. The central-minus-local difference is about -0.11 pp, with a 95% CI of [-0.49, +0.27] pp. Industry, size, market position, and specific policy mandates appear more informative than the central/local split.

5.7 Financial Subsidy: About CNY 682.6 Billion in 2025

The firm-level subsidy rate becomes an amount after multiplying by book assets and summing across firms:

$$\widehat{FS}_\tau = \sum_i \widehat{FSR}_{i,\tau} \times A_{i,\tau}^{SOE}. \quad (12)$$

The resulting financial subsidy is reported in CNY 100 million.

Table 8: Estimated Financial Subsidies by Year (CNY 100 Million)

Year	All SOEs	CSI 300	CSI 500	CSI 1000	Non-index
2019	5,557	3,528	850	492	687
2020	10,059	6,273	1,919	494	1,373
2021	3,595	1,364	1,257	233	741
2022	3,167	761	1,719	736	-48
2023	6,375	2,366	2,343	1,262	404
2024	5,626	315	2,117	1,043	2,151
2025	6,826	1,724	1,826	1,115	2,161

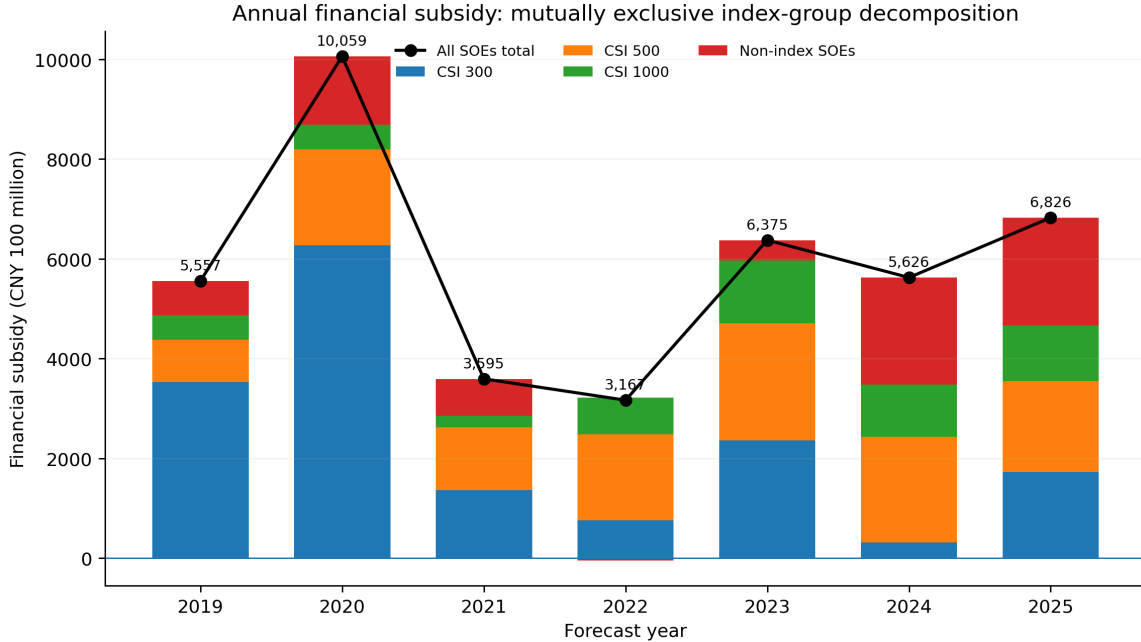


Figure 5: Annual financial subsidy and its mutually exclusive index-group decomposition. The line shows the all-SOE total; stacked bars show CSI 300, CSI 500, CSI 1000, and non-index SOEs.

Under the raw-outcome specification, the 2025 subsidy is CNY 682.6 billion, and the simple

nominal sum for 2019–2025 is CNY 4.12 trillion. In 2025, Good or Boundary firms account for 83.6% of the subsidy, while the Extrapolation region accounts for 16.4%.

The rate and the amount answer different questions. The rate describes the return gap for a typical firm; the amount weights each firm’s gap by its asset base. The 2025 CSI 300 rate is close to zero (-0.04 pp), yet its aggregate amount is about CNY 172.4 billion because a small number of large firms can materially affect the asset-weighted sum.

5.8 Robustness Checks

5.8.1 Outcome Definition: The 1.5 pp Result Is Not Driven by Tail Observations

The full-market rate of about 1.5 pp survives symmetric outcome winsorization. The main specification uses raw EBI/A. For the robustness check, each forecast year’s P5 and P95 thresholds come only from the historical private-firm training sample, and the same thresholds apply to both the training target and that year’s SOE realizations.

Table 9: Robustness to Outcome Definition: Multiyear Financial Subsidy Rate, 2019–2025 (pp)

Sample	Raw–Raw	95% CI	Symmetric winsorization	95% CI
All SOEs	+1.45	[+1.25,+1.64]	+1.50	[+1.36,+1.64]
CSI 300	-0.07	[-0.69,+0.56]	+0.24	[-0.26,+0.74]
CSI 500	+1.34	[+1.04,+1.65]	+1.54	[+1.27,+1.80]
CSI 1000	+1.39	[+1.10,+1.71]	+1.40	[+1.17,+1.63]

The full-sample rate is +1.45 pp under the raw specification and +1.50 pp under symmetric winsorization, a difference of 0.05 pp. Symmetric winsorization affects about 8.4% of SOE firm-years, mostly in the lower tail of EBI/A. An earlier asymmetric specification—winsorized prediction minus raw realization—produced about 2.02 pp, but the prediction and realization used different outcome definitions, so it is not retained as a formal specification.

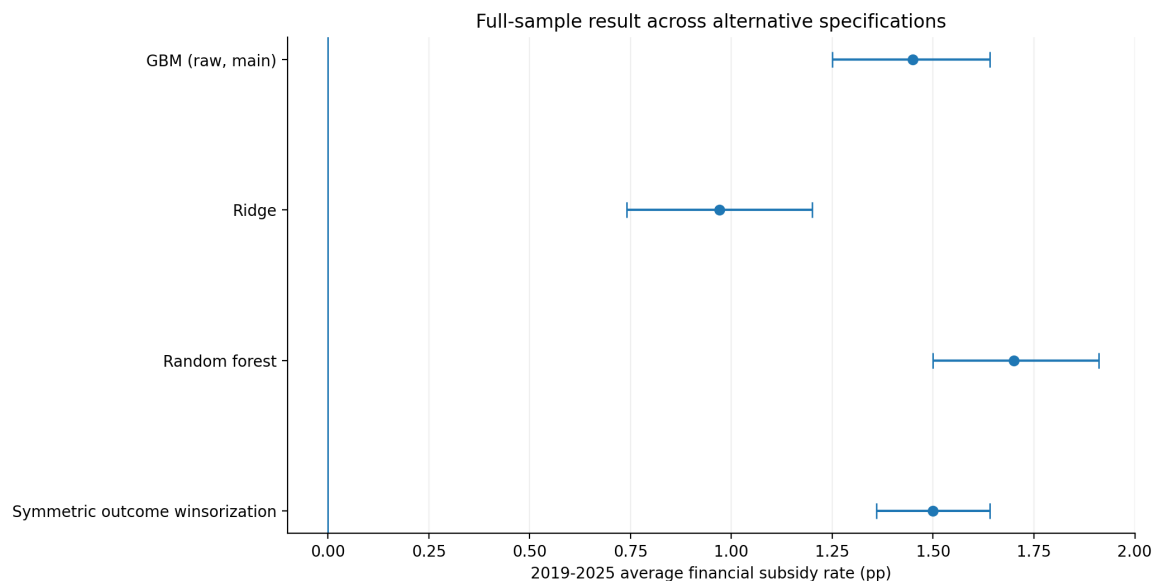


Figure 6: Full-sample financial subsidy rate across alternative specifications. Horizontal bars report 95% confidence intervals.

5.8.2 Common Support: The Full-Sample Subsidy Rate Is Almost Unchanged in Comparable Samples

Common support (CS) is Good + Boundary, or $d^{(5)} \leq P95_T$. A persistent-good firm spends at least 80% of its observed years in Good support. All three multiyear estimates use 2,000 firm-level cluster-bootstrap draws.

Table 10: Multiyear Common-Support Results, 2019–2025

Sample	Full rate	CS rate	Persistent-good rate	Mean Good%	Mean CS%
All SOEs	+1.45 [+1.25,+1.64]	+1.42 [+1.22,+1.62]	+1.41 [+1.18,+1.64]	79.9	90.2
CSI 300	-0.07 [-0.69,+0.56]	-0.12 [-0.86,+0.63]	+0.67 [-0.08,+1.46]	56.4	77.1
CSI 500	+1.34 [+1.04,+1.65]	+1.17 [+0.84,+1.50]	+1.26 [+0.77,+1.80]	70.4	85.7
CSI 1000	+1.39 [+1.10,+1.71]	+1.34 [+1.03,+1.68]	+1.65 [+1.20,+2.14]	80.4	91.1

Note: Each rate cell reports the point estimate on the first line and the 95% confidence interval in brackets on the second. CS = Good + Boundary ($d^{(5)} \leq P95_T$). Persistent-good uses firms with $GoodShare_i \geq 0.8$. Mean Good% and Mean CS% are firm-level averages of GoodShare and CSShare. Confidence intervals use 2,000 firm-level cluster-bootstrap draws (seed=42).

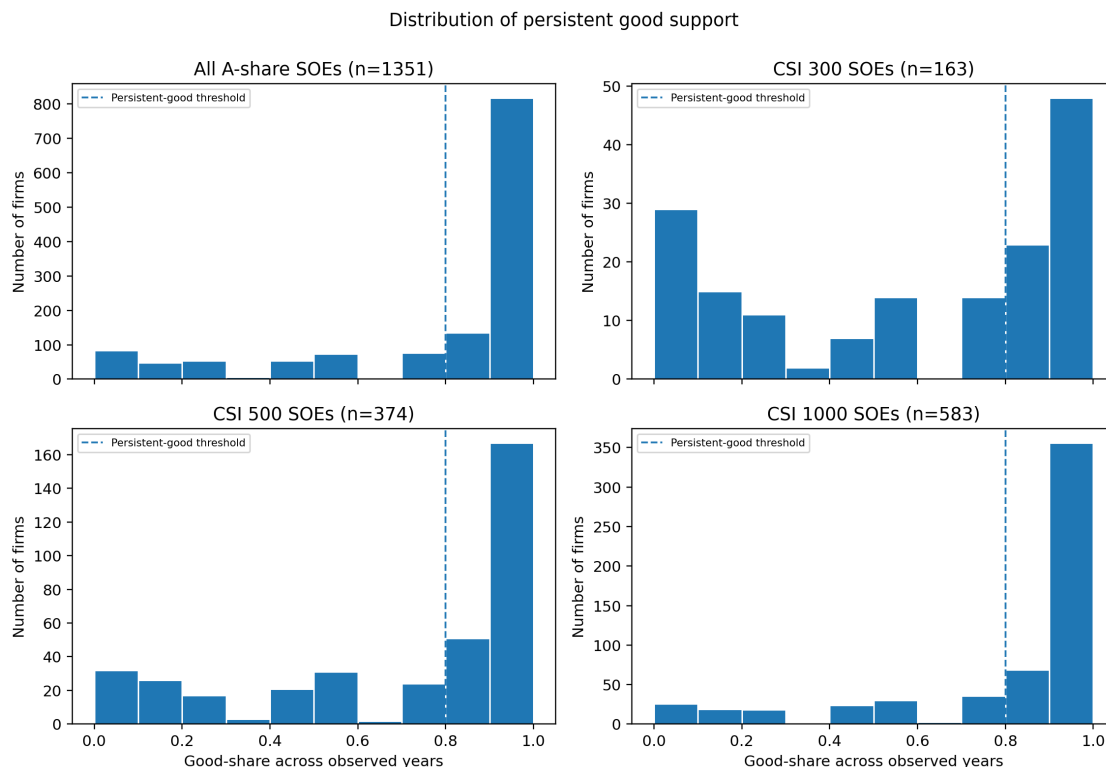


Figure 7: Distribution of firm-level Good-support shares. The dashed line marks the 0.8 threshold used to define persistent-good firms.

For all A-share SOEs, the Full (+1.45 pp), CS (+1.42 pp), and persistent-good (+1.41 pp) estimates are nearly identical, and all 95% confidence intervals exclude zero. Mean CSShare is 90.2%. CSI 500 and CSI 1000 also remain positive after support restrictions: CS rates are +1.17 pp and +1.34 pp, and persistent-good rates are +1.26 pp and +1.65 pp, with all confidence intervals above zero.

CSI 300 remains different. The full rate (-0.07 pp) and CS rate (-0.12 pp) are close to zero and insignificant. The persistent-good estimate is +0.67 pp, but its 95% CI [-0.08,+1.46] includes zero. Mean GoodShare is 56.4% and Mean CSShare is 77.1%, both below the other index groups. Under GBM, positive rates for the full market and CSI 500/1000 survive support restrictions; CSI 300 has no significantly positive rate under the Full, CS, or persistent-good definitions.

In 2025, Good firms contribute about CNY 325.3 billion (47.7%), Boundary firms CNY 245.2 billion (35.9%), and Extrapolation firms CNY 112.2 billion (16.4%) to the CNY 682.6 billion all-SOE amount. Good + Boundary firms contribute about CNY 570.4 billion, or 83.6% of the total. Both the equal-weighted rate and the asset-weighted amount are therefore generated mainly by firms with private-sector common support.

Where the 2025 financial subsidy amount comes from

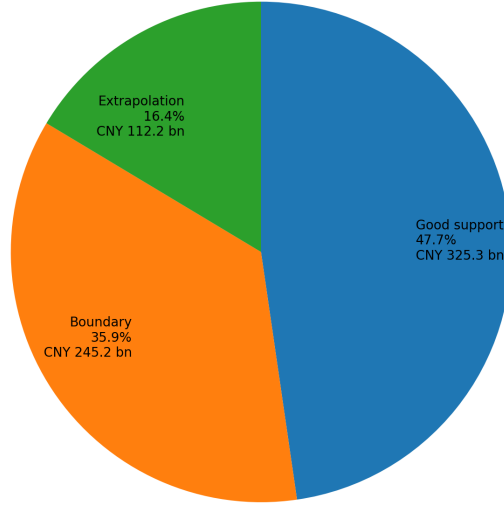


Figure 8: Common-support composition of the 2025 financial subsidy for all A-share SOEs.

Table 11: Common Support and Financial-Subsidy Decomposition in 2025

Sample	CS%	Extrapolation%	Full rate	CS rate	Subsidy (CNY 100m)	Extrapolation share
All SOEs	88.8	11.2	+1.53 pp	+1.45 pp	6,826	16.4%
CSI 300	73.5	26.5	-0.04 pp	-0.12 pp	1,724	-24.5%
CSI 500	82.0	18.0	+1.22 pp	+1.09 pp	1,826	26.0%
CSI 1000	88.6	11.4	+1.22 pp	+1.13 pp	1,115	27.2%

Note: CS = Good + Boundary. Extrapolation share is the subsidy amount in the extrapolation region divided by the total subsidy amount. Financial subsidy is a signed sum of subsidy rate $\times$ book assets. The negative CSI 300 extrapolation share means that extrapolation-region firms have a negative aggregate subsidy in 2025 and therefore offset part of the group's total.

5.8.3 Model Class: Robust Full-Sample Direction, Greater Sensitivity in Subsamples

Changing the model changes the magnitude of the full-sample rate but not its positive direction. Multiyear rates are +1.45 pp for GBM, +0.97 pp for Ridge, and +1.70 pp for RF; all 95% confidence intervals are above zero. The corresponding common-support and persistent-good estimates are also significantly positive.

Index subsamples are less stable across model classes. For CSI 500, GBM and RF yield significantly positive multiyear rates, while Ridge gives +0.11 pp with a confidence interval that includes zero. For CSI 300, GBM is close to zero while Ridge and RF give negative estimates. For CSI 1000, all three models are positive, with magnitudes ranging from about +0.50 pp to +1.55 pp. Model replacement therefore supports a robust positive full-market direction but a more model-sensitive interpretation of smaller subsamples.

5.8.4 Inference: Full-Sample Confidence Intervals Remain Above Zero

Firm-level cluster bootstrap leaves the full-sample inference unchanged. The GBM 95% confidence intervals for Full, CS, and persistent-good estimates are $[+1.25, +1.64]$, $[+1.22, +1.62]$, and $[+1.18, +1.64]$. Ridge and RF multiyear intervals are $[+0.74, +1.20]$ and $[+1.50, +1.91]$. All intervals use 2,000 firm-level cluster-bootstrap draws (seed=42).

5.8.5 Benchmark Validity: No Systematic Overprediction in Private Firms

Private-firm OOF residuals provide a placebo for systematic overprediction. The pooled residual (Predicted - Actual) is +0.002 pp across 26,464 firm-years from 3,344 private firms. The firm-clustered p -value is 0.979, with a 95% CI of $[-0.14, +0.14]$ pp. A benchmark that systematically overpredicted returns should also produce positive residuals for private firms; the placebo does not show that pattern.

GBM also matches the scale of realized private-firm returns well. Its pooled raw-outcome OOF calibration slope is 0.98, with an intercept of about 0.001 and $R^2 = 0.239$. RF's slope is about 1.39, indicating greater shrinkage toward the mean. Ridge has a slope of about 0.97 but lower explanatory power than GBM. Out-of-time fit, RMSE, and calibration jointly support GBM as the main model.

6 Conclusion

Chinese A-share listed SOEs earn lower asset returns than the conditional private-sector benchmark on average. Under the GBM specification, the 2019–2025 financial subsidy rate is 1.45 pp, with annual estimates positive and statistically significant in every year. Restricting the sample to common support produces 1.42 pp, and persistent-good firms produce 1.41 pp. Ridge, RF, symmetric outcome winsorization, and the private-firm OOF placebo all preserve the positive full-market conclusion.

The average masks substantial heterogeneity. CSI 500 and CSI 1000 SOEs have multiyear rates of +1.34 pp and +1.39 pp, while CSI 300 is close to zero under the main GBM specification. Industry rates differ sharply and move over time, whereas central and local SOEs have similar averages. Multiplying firm-level rates by assets gives a 2025 subsidy of about CNY 682.6 billion, with 83.6% generated inside common support.

The estimated scale is also consistent with prior Chinese financing evidence. Our annual 0.74–1.88 pp range overlaps debt-side estimates, while the 2019 and 2023 aggregate amounts are of the same order as independent below-market-credit estimates. These comparisons benchmark economic scale rather than equality of estimands.

The measurement design also has limits. The subsidy rate is a conditional return gap rather than the causal effect of privatization, and the amount is an asset-return opportunity-cost measure rather than a decomposition of specific support channels. Common support depends on observed covariates, the ownership and accounting fields impose measurement constraints, the bootstrap

omits first-stage retraining uncertainty, and the sample covers listed rather than all Chinese SOEs. Those limits are most consequential for the largest firms, especially CSI 300 SOEs, where private-sector overlap is weaker and model sensitivity is greater.

The main policy implication concerns measurement rather than mechanism. Asset-side EBI/A puts debt and equity support on a common scale, while a strictly out-of-time conditional benchmark allows the comparison return to vary with firm characteristics. Future work can combine the benchmark with richer product-market, governance, geography, and policy-mandate data; extend it to unlisted and non-mainland listings; and use quasi-natural experiments in credit policy, SOE reform, regulation, or governance to identify why persistent subsidy rates arise and who ultimately benefits from them.

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Appendix A. Additional Common-Support Diagnostic

The multivariate 5-NN diagnostic provides the main common-support classification. Figure A1 adds a one-dimensional view of the most visible mismatch, firm size. The figure is descriptive rather than the final support rule, but it helps explain why private-sector benchmarks are most difficult to construct for CSI 300 SOEs.

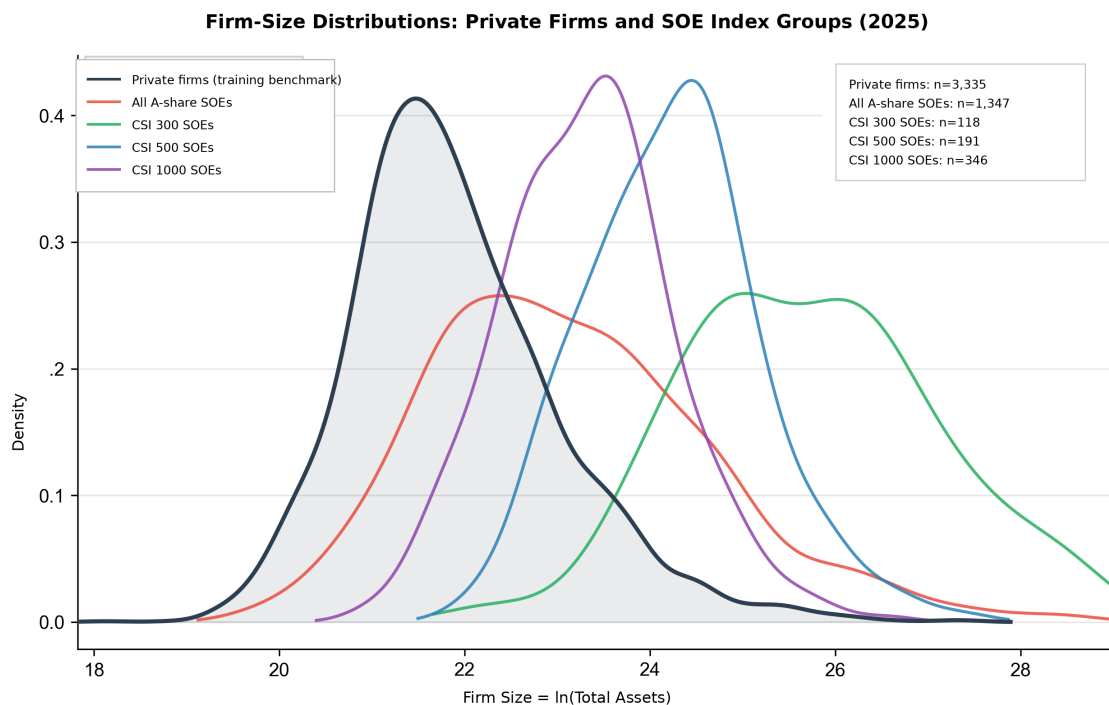


Figure A1: Firm-size distributions for private firms and SOE index groups in the screened 2025 sample.